TEXORA · LECTURE SHEET

Introduction to Dyeing and Printing

First principles of coloration: what dyes and pigments are and how they differ, the classes each divides into and the fibres they serve, how dyes and pigments are applied, and the distinction between dyeing a whole material and printing a design onto it.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 8 · LEVEL 2 TERM 2

1. Colourants: dyes and pigments defined

A **dye** is a coloured substance that is applied to a textile from a solution or fine dispersion, is absorbed *into* the fibre, and has a chemical **affinity** for that fibre — it becomes part of the material at the molecular level. A **pigment** is a coloured substance that is **insoluble** in water and has **no affinity** for any fibre; it can only be fixed *onto* the surface by a polymeric **binder** film that glues it in place. Both are called colourants; nearly everything about their behaviour follows from this solubility/affinity distinction.

Colour in a dye molecule comes from a conjugated system of alternating double bonds — the **chromophore** (e.g. azo –N=N–, anthraquinone, phthalocyanine structures) — modified by electron-donating/withdrawing groups called **auxochromes** (–NH_2, –OH, –SO_3H…) that shift the shade and provide the fibre-binding and solubilising sites. Pigments use many of the same chromophores, engineered into insoluble crystal lattices.

2. Dyes versus pigments

PROPERTY	DYES	PIGMENTS
Solubility	Soluble in water, or made soluble during application (vat, sulphur)	Insoluble in water and solvents
Affinity for fibre	Yes — specific to fibre classes	None — fibre-independent
Site of colour	Inside the fibre (molecular dispersion)	On the fibre surface, in a binder film
Application medium	Dye bath (exhaustion) or padding	Print paste or pad liquor containing binder

Fixation mechanism	Adsorption, ionic/covalent bonding, or precipitation inside fibre	Binder polymerisation/curing around particles
Handle of fabric	Unchanged	Can stiffen (depends on binder and coverage)
Fastness profile	Class-dependent; wash fastness often high	Excellent light fastness; rub fastness limited by binder
Typical use	Dyeing of all fibre types	Printing (main use), mass pigmentation, coatings

3. Classes of dyes

Dyes are classified by chemical structure and — more usefully in the dyehouse — by **method of application**, which ties each class to the fibres it can colour:

CLASS	FIBRES	FIXATION PRINCIPLE / KEY FEATURE
Direct	Cellulosics	Substantivity from long planar molecules (H-bonding, van der Waals); simple exhaust dyeing with salt; moderate wash fastness
Reactive	Cellulosics (also wool, silk, nylon types)	Forms a **covalent bond** with the fibre; brightest shades and best wash fastness on cotton
Vat	Cellulosics	Insoluble pigment-like dye reduced to soluble leuco form, absorbed, then oxidised back inside the fibre; outstanding wash/light fastness (indigo)
Sulphur	Cellulosics	Similar reduce–absorb–oxidise cycle; cheap, dull dark shades; tendering risk on storage
Azoic	Cellulosics	Colour formed *inside* the fibre by coupling two soluble components (naphthol + diazonium salt)
Acid	Wool, silk, nylon	Anionic dye attracted to protonated amino sites under acid conditions
Basic (cationic)	Acrylic (main modern use)	Cationic dye bonds to anionic sites in acrylic; brilliant shades
Disperse	Polyester, acetate, synthetics	Non-ionic, sparingly soluble; dissolves molecularly into the hydrophobic fibre at high temperature

Mordant	Wool mainly	Dye complexed with a metal ion (e.g. Cr) anchored to the fibre

4. Classes of pigments

- **Inorganic pigments:** titanium dioxide (white), carbon black, iron oxides (red, yellow, brown, black), ultramarine, chromium oxide green. Very high light and heat stability, opaque, generally dull.
- **Organic pigments:** insoluble forms of chromophores — phthalocyanine blue and green, azo pigments (reds, yellows), quinacridones. Brighter shades, the mainstay of textile printing.

Because pigments have no affinity, all pigment coloration needs a binder system, thickener (for printing) and curing — the pigment, not the fibre, dictates nothing; one pigment paste colours every fibre and blend equally, which is precisely why printing is dominated by pigments.

5. Dyeing: definition and mechanism

Dyeing is the process of applying colour uniformly over the whole material (fibre, yarn, fabric or garment) by immersion in a dye bath or by padding, so that dye transfers from the bath into the fibre. It proceeds in four stages:

1. **Migration/exhaustion:** dye molecules move through the bath to the fibre surface (assisted by temperature, salt, pH according to class).
2. **Adsorption:** dye accumulates on the fibre surface at active sites.
3. **Diffusion:** dye migrates from the surface into the fibre interior — slow, temperature-driven, and the rate-controlling stage.
4. **Fixation:** the dye is secured — by covalent reaction (reactive), oxidation to the insoluble form inside the fibre (vat, sulphur), ionic bonding (acid, basic) or physical entrapment (disperse).

After dyeing, unfixed dye is washed off (soaping) — the step that decides real-world wash fastness.

6. Printing: definition

Printing is localised coloration: a thickened print paste containing dye or pigment is applied to selected areas of the fabric in a design (by screen, roller or digital inkjet), then fixed by steaming/curing and washed. The thickener (starch, alginate, synthetic thickeners) holds the colour exactly where placed and prevents it spreading by capillary action into the design's edges. Printing styles include **direct** (paste printed on white/prepared ground), **resist** (a resist paste prevents dyeing of printed areas) and **discharge** (a reducing paste destroys ground dye to leave a pattern).

7. Dyeing versus printing

ASPECT	DYEING	PRINTING
Area coloured	Whole material, uniformly	Selected areas, in a design
Medium	Thin dye bath (liquor ratio matters)	Thick print paste (thickener essential)
Colourant concentration	Low (typically 0.1–5% owf)	High, localised
Face/back appearance	Same both sides	Design side stronger; back usually paler
Fixation	In the bath, during exhaustion	Afterwards — steaming, curing or baking
Water/energy use	High (large baths, washing)	Lower overall (little washing for pigment prints)
Design capability	Plain shades (plus yarn-dyed patterns)	Multi-colour designs, photographs (digital)

8. Glossary — key terms

- **Chromophore:** the conjugated group responsible for a colourant's colour.
- **Auxochrome:** a group modifying shade and giving solubility/affinity.
- **Affinity (substantivity):** the attraction between dye and fibre driving exhaustion.
- **Binder:** the polymer film former that fixes pigment to the surface.
- **Leuco form:** the reduced, soluble, nearly colourless form of vat/sulphur dyes during application.
- **Exhaustion:** the proportion of bath dye transferred to the fibre.

9. Key points

- Dyes dissolve and bond inside specific fibres; pigments are insoluble, affinity-free, and need a binder on the surface.
- Dye classes map to fibres: reactive/direct/vat/sulphur/azoic → cellulosics; acid → protein/polyamide; basic → acrylic; disperse → polyester.
- Dyeing = whole-material coloration through migration, adsorption, diffusion, fixation. Printing = localised coloration from a thickened paste.
- The thickener and the after-fixation step are what make printing a different technology, not merely dyeing with a stencil.

10. Summary

Coloration rests on two kinds of colourant. Dyes enter the fibre and are held by real chemical or physical affinity — hence the class system matching dye to fibre. Pigments sit on the surface in a binder film and therefore colour anything. Dyeing colours material uniformly from a bath; printing applies a thickened colour locally to build designs. Everything in the later coloration courses — dyeing machinery, fastness, print technology — is an elaboration of these four definitions and two comparison tables.

11. Practice

Answer in your own words; brief model answers follow.

1. Indigo in a vat bath is almost colourless and soluble, yet the finished jeans are blue and wash-fast. Explain with the vat-dye mechanism.
2. Why can one pigment print paste colour a polyester/cotton blend evenly, while no single dye can?
3. Give four differences between dyeing and printing.
4. A printer wants sharp outlines on an absorbent cotton fabric. Why is thickener indispensable in the paste, and what goes wrong without it?
5. Choose a dye class for each: (a) bright red cotton T-shirt with high wash fastness; (b) navy polyester sportswear; (c) brilliant violet acrylic sweater. Justify briefly.

Answer key

1. Indigo is reduced in the alkaline vat to its soluble leuco form, which the cotton absorbs; on exposure to air it oxidises back to insoluble blue indigo *inside* the fibre, mechanically locked in — hence fastness (and surface-concentrated dye, hence the characteristic fading of denim).
2. Pigments have no fibre affinity and are fixed by a binder film on the surface, so both fibres are coated identically. Dyes work by fibre-specific affinity: a cotton dye ignores polyester and vice versa, so blends need two dye classes in one or two baths.
3. Any four from the table: area coloured (whole vs design); medium (thin bath vs thick paste); colourant concentration; face/back equality; fixation timing (in bath vs after steaming/curing); water use; design capability.
4. The thickener gives the paste a high viscosity so it stays exactly where the screen places it instead of wicking along yarns and capillaries into unprinted areas. Without it the colour bleeds — outlines blur and fine detail is lost.
5. (a) Reactive — covalent bonding gives bright shades with the best wash fastness on cotton. (b) Disperse — the only class that dissolves into hydrophobic polyester. (c) Basic (cationic) — bonds to

acrylic's anionic sites with brilliant shades.
